Community Mental-Wellbeing Signal Tracker

TECHNICAL REPORT

Domain / Focus: Sri Lanka Regional Stressor Monitoring	**Repository:** nonimml/community-wellbeing-tracker
Deployment Mode: Synthetic Fallback / Demonstration	**Privacy Rule:** Aggregate-Only (k >= 10 Suppression)

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1. Executive Overview

This report breaks down the technical execution, system architecture, technology selections, and baseline evaluation for the Community Mental-Wellbeing Signal Tracker project. Developed as an academic data science prototype centered on Sri Lanka, the application offers regional situational awareness and early-warning signal monitoring across 25 districts and 9 provinces. The primary objective is delivering actionable insights to wellbeing programme coordinators while strictly guaranteeing individual privacy through minimum-group cell suppression (k = 10).

2. System Architecture & Module Pipeline

The application follows a clean, decoupled design where raw inputs, transformation pipelines, analytical aggregations, and visual presentation layers remain distinctly separated:

Layer	Modules	Responsibility & Functional Scope
Data Ingestion & Synthetic Engine	database.py generate_data.py	Manages database connections, data sanitization, and fallback synthetic generation across 25 Sri Lankan districts.
Privacy & Governance	privacy.py geography.py	Applies k=10 threshold cell suppression and maps district-level inputs cleanly to Sri Lanka's 9 provinces.
Analytics Engine	analysis.py	Computes weekly aggregations, Z-score anomaly indicators for early warning, and statistical confidence intervals (CIs).
User Presentation	app.py	Multi-page Streamlit web application providing interactive filters, trend charts, and evaluation dashboards.

3. Tech Stack Justification

The stack was selected to prioritize high development velocity, maintain statistical precision, and avoid unnecessary server infrastructure infrastructure burdens:

- **Python 3.10+:** Provides a robust data science ecosystem and maintainable codebase.
- **Streamlit:** Allowed rapid creation of multi-page interactive dashboards directly from Python without frontend framework complexity.
- **Pandas & NumPy:** Handles vectorized calculations, data cleaning, and fast aggregation over thousands of survey entries.
- **Plotly Express:** Renders dynamic, interactive visualizations including trend bands, hover tooltips, and bar charts.

- **Supabase / PLpgSQL:** Provides PostgreSQL database integration and geographic SQL migration scripts.

4. Evaluation Against Proposal

Proposed Feature	Status	Honest Technical Assessment
5 Core Stressor Themes	**Delivered**	Successfully tracks Job/Money, Exams/Studies, Loneliness, Family, and Other across regions.
Privacy Safeguards (k >= 10)	**Delivered**	privacy.py active cell suppression prevents displaying groups smaller than 10 responses.
Sri Lanka Regional Mapping	**Delivered**	Full 25-district data sampling structured into 9 provinces via geography.py.
Early Warning System	**Enhanced**	Integrated Z-score surge alerts, trend Confidence Intervals (CIs), and an evaluation view in app.py.
Live NLP Classification	**Synthetic Fallback**	Uses simulated category labels rather than live human text classification to maintain strict ethical safety boundaries.

5. Next Steps & Recommendations

- Obtain formal ethics board (IRB) approval prior to ingesting real community survey data.
- Implement fine-tuned Sinhala and Tamil NLP models to support local language feedback.
- Explore mathematical Differential Privacy (e.g. Laplace noise) as an enhancement beyond minimum k-anonymity cell suppression.